

Supplementary Material

Article Title: Feature-Oriented Model-Based GUI Test Generation for Mobile Applications

Authors: Ali Asghar Yarifard, Saeed Araban, Samad Paydar, Vahid Garousi

Date: July 18, 2026

Document ID: 2026-V1

1. Overview

This document serves as the official Supplementary Material for the research paper titled “*Feature-Oriented Model-Based GUI Test Generation for Mobile Applications*”. To manage the complexity of the EM model presented in the main manuscript, we utilize a standardized identifier system for thhe GUI event handlers used in the EM model. In this system, each GUI event handler is represented in the visual model by a unique ID (e.g., MH_01, SH_01). This document provides the full functional mapping for these identifiers. The purpose of this document is to improve readability in the main manuscript while preserving full traceability of all GUI event handler definitions.

2. Identifier Naming Convention

To ensure clarity and modularity within the EM model, each GUI event handler is assigned a unique alphanumeric identifier. These identifiers follow a standardized format consisting of two parts which are as follows:

- Prefix (Context):** a page prefix indicating the GUI context: MH (MainPage), CH (ContactCreatePage), SH (SettingPage), EH (EmailCategoryPage), ECH (EditContactPage), NH (NumberCategoryPage), PH (PermissionPage).
- Suffix (Number):** a sequence number identifying the event handler (e.g., 01, 02).

3. Table S1: Mapping of GUI Event Handlers

The following table provides the exhaustive mapping between the visual identifiers used in the EM model and their corresponding functional definitions within the application.

ID	Handler Name	Page Context	Functional Description / Transition Target
MH_01	OpenDeleteContactDialog	MainPage	Opens the dialog to delete the selected contatc.
MH_02	ConfirmDeleteContact	MainPage	Confirms and executes contact deletion logic.
MH_03	CancelDeleteContact	MainPage	Dismisses the deletion dialog and reverts to idle state.
MH_04	OpenContactCreatePage	MainPage	Navigates to the Contact Create view.
MH_05	OpenSettings	MainPage	Navigates to the Setting view.
MH_06	OpenEditContactPage	MainPage	Navigates to the Edit Contact view.
CH_01	SaveContact	ContactCreatePage	Validates input and stores the new contact entity.
CH_02	OpenAddNumberDialog	ContactCreatePage	Opens the dialog to add a new phone number.
CH_03	OpenAddEmailDialog	ContactCreatePage	Opens the dialog to add a new Email.
CH_04	OpenDeleteNumberDialog	ContactCreatePage	Opens the dialog to delete the selected phone number.
CH_05	ConfirmDeleteNumber	ContactCreatePage	Confirms and executes phone number deletion logic.
CH_06	CancelDeleteNumber	ContactCreatePage	Dismisses the deletion dialog and reverts to idle state.
SH_01	OpenEmailCategoryPage	SettingPage	Navigates to the email category configuration.
SH_02	OpenPermissionPage	SettingPage	Navigates to the Permission management view.
SH_03	OpenNumberCategoryPage	SettingPage	Navigates to the number category configuration.
ECH_01	OpenEditContactPage	EditContactPage	Navigates to the edit contact view.

ID	Handler Name	Page Context	Functional Description / Transition Target
ECH_02	OpenCallApp	EditContactPage	Navigates to calling App to call the selected phone number
ECH_03	OpenMailApp	EditContactPage	Navigates to email App to send an email to the selected email address
EH_01	OpenAddEmailCatrgoryDialog	EmailCategoryPage	Opens the dialog to add a new category for email address.
EH_02	ConfirmAddEmailCategory	EmailCategoryPage	Confirms and executes email category addition logic.
EH_03	CancelAddEmailCategory	EmailCategoryPage	Dismisses the add email category dialog and reverts to idle state.
EH_04	OpenDeleteEmailCategoryDialog	EmailCategoryPage	Opens the dialog to delete the selected category for email address.
EH_05	ConfirmDeleteEmailCategory	EmailCategoryPage	Confirms and executes email category deletion logic.
EH_06	CancelDeleteEmailCategory	EmailCategoryPage	Dismisses the delte email category dialog and reverts to idle state.
PH_01	SetPhonePermission	PermissionPage	Configures calling permission.
PH_02	SetCameraPermission	PermissionPage	Configures camera permission.
NH_01	OpenAddNumberCatrgoryDialog	NumberCategoryPage	Opens the dialog to add a new category for phone number.
NH_02	ConfirmAddNumberCategory	NumberCategoryPage	Creates a new number category entry.
NH_03	CancelAddNumberCategory	NumberCategoryPage	Dismisses the phone number category adding dialog and reverts to idle state.
NH_04	OpenDeleteNumberCategoryDialog	NumberCategoryPage	Opens the dialog to delete the selected category for phone number.
NH_05	ConfirmDeleteNumberCategory	NumberCategoryPage	Removes an existing number category entry.
NH_06	CancelDeleteNumberCategory	NumberCategoryPage	Dismisses the phone number category deletion dialog and reverts to idle state.

3. Contact Information

For further technical inquiries regarding the EM model or access to the underlying XML/JSON artifacts, please contact the corresponding author via the email address provided in the main manuscript.